SILICON LABS
CONNECTED INTELLIGENCE

Bluetooth® Channel Sounding

The Leader in IoT Wireless Connectivity

100%
IoT Focused

amazon sidewalk · Bluetooth · matter · Multiprotocol · Proprietary · THREAD · Wi Fi · Wi SUN · zigbee · Z WAVE

Breadth and Depth of Wireless IoT Protocols

#1
Share in Mesh

1st
To Market with Multiprotocol, BLE Mesh, BLE 5.1

Innovation
Performance, Power, CoEx, Modules, SecureVault™

ember	ENERGY micro	bluegiga	telegesis	Micriμm	ZENTRI	Z WAVE	REDPINE SIGNALS
2012	2013	2015	2015	2016	2017	2018	2020
Software ZigBee SoC	Low-power 32-bit MCUs	BT Smart Modules	ZigBee/Thread Modules	Software RTOS	Cloud Connected Wi-Fi	Smart Home Protocol	Ultra Low Power Wi-Fi

A Complete and Robust Bluetooth® Offering

DESIGN | DEVELOPMENT | DEPLOYMENT

Developer Kits

- Exploratory kits with standard connectors
- Full featured kits with LCD, Power Measurement, etc.
- Demo kits w/ built in sensors

200+ OOB demos and sample applications

Cutting-edge BT Stack

- Bluetooth LE and Mesh
- Supporting most BLE features
- Certified against latest spec

~1B BLE units deployed

Mobile Apps

- BLE scanner/advertiser
- Demos
- IOP testing with smartphones
- Bluetooth Mesh ADK and App

100+ Mobile phones with next-gen IOP testing

ICs and Modules

- QFN, CSP ICs
- PCB and SiP modules
- Antenna Design (int/ext)
- Optimized size
- Pre-certified modules

1st in the market with PSA L3 certification

Superior Developer Tools

- GATT Configurator
- Bluetooth NCP Commander
- Network Analyzer
- AoA Analyzer
- Energy Profiler

~1M Packets transmitted daily for Mesh endurance testing

Reliability Testing, IOP

- Large Scale Test Network
- IOP testing
- Regression testing & CI
- Throughput and range test

7,000+ functional tests for Bluetooth and mesh

2025

- SiBG301, first Series 3 BLE product, announced
- BG29 announced
- BG22L and 24L launched to boost BLE portfolio
- Channel Sounding Dev Kit Launched

2024

- BT 6.0 spec adoption
- ESL Large-Scale Test Network (~1000+ nodes)
- BG26 announced
- Channel Sounding support available

2023

- BT 5.4 spec announced
- SiLabs adoption of 5.4
- BG27 released with 5.4
- Mesh Large Scale Network

2022

- BG24 released w/ 5.3
- Mesh 1.1 announced
- Series 2 SoCs support mesh 1.1

2021

- First PSA L3 SoC
- BT 5.3 spec adoption
- Enabled CPMS integration
- Series 2 SoCs support 5.3

2015

Shipped 1st BLE SoC & module

Why Bluetooth® Channel Sounding?

THE CHALLENGE:

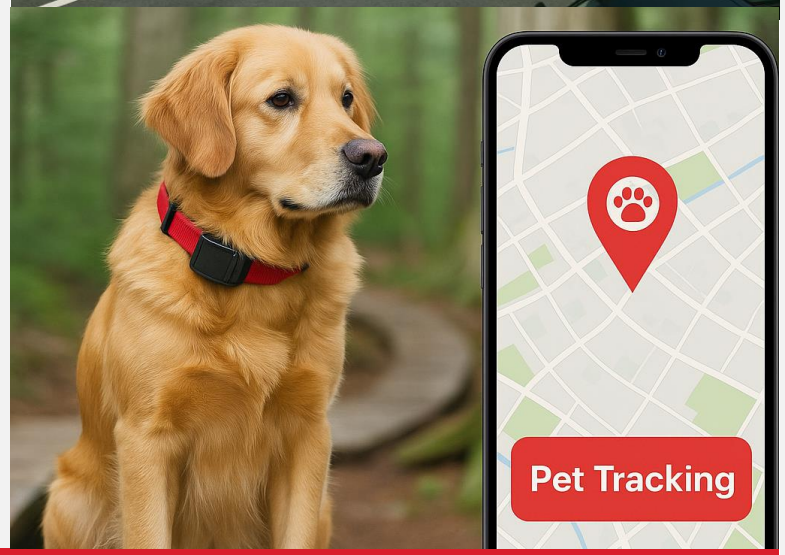
IOT applications need 'spatial' awareness to be more secure, reliable, and responsive

- Earlier Bluetooth LE versions lack native support for precise ranging
- RSSI-based ranging is noisy and unreliable in real-world use
- Direction Finding needs complex antennas, adding cost & complexity
- UWB is accurate but often too costly, power-hungry, and bulky for IoT

THE OPPORTUNITY:

Channel Sounding for Bluetooth

- Standardized approach for accurate, secure ranging in Bluetooth 6.0
- Enables sub-meter accuracy with robust performance, even in NLOS
- Works with single or dual antenna setups - flexible for different form factors
- More cost and power-efficient than UWB; requires minimal external components



Built on existing Bluetooth infrastructure, simplifying adoption and ecosystem integration

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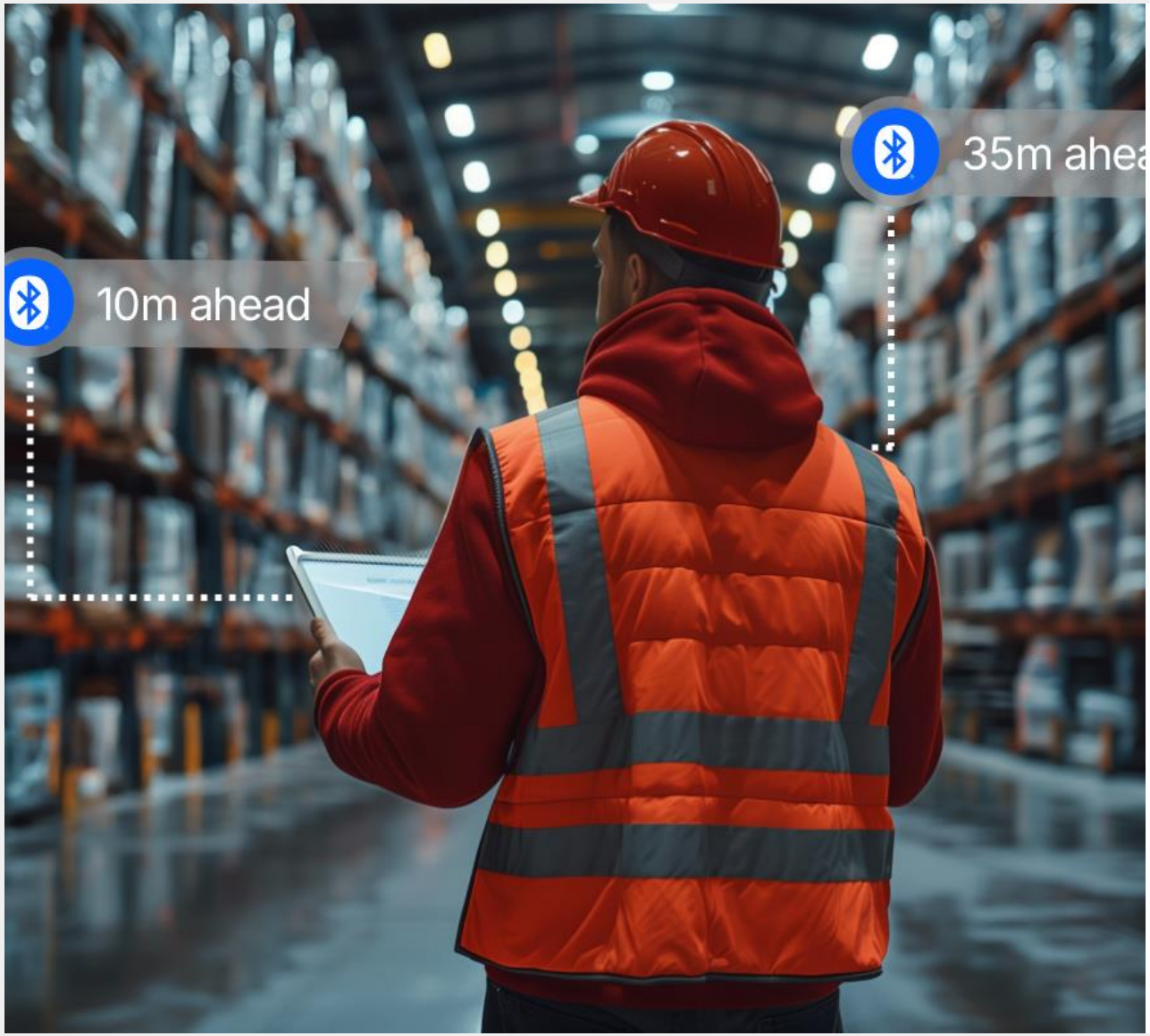
Driving Innovation with Bluetooth® Channel Sounding



Enables high-accuracy distance measurement

- Resolve distance between two connected Bluetooth LE devices
- Support for phased based ranging technique and Time of Flight
- Multi-antenna supported for optimized Non-Line-Of-Sight ranging

Standardized for interoperability and reliability

- Ensures compatibility with a wide range of devices
- Device performance ensured through Bluetooth qualification

Cost effective and efficient

- Unlike UWB based solutions, potential for single-chip ranging
- Simplifies layout design, saving board space and cost

Enhanced Security

- Reduces likelihood of spoofing and man in the middle attacks
- Deterministic random bit generator randomizes the bit stream, payloads in tone extensions, antenna path selection and more
- NADM, or normalized attack detector metric, communicates likelihood of bit manipulation in RTT
- Mode3 (RTT+PBR) allows cross-check of data from PBR with RTT to prevent man-in-the-middle (MITM) attack

Flexible architecture

- Software stack supports device operating in NCP or SoC mode

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Bluetooth Channel Sounding - Target Markets & Use Cases

PROXIMITY AWARENESS

Door locks

Keyless entry

Building access systems

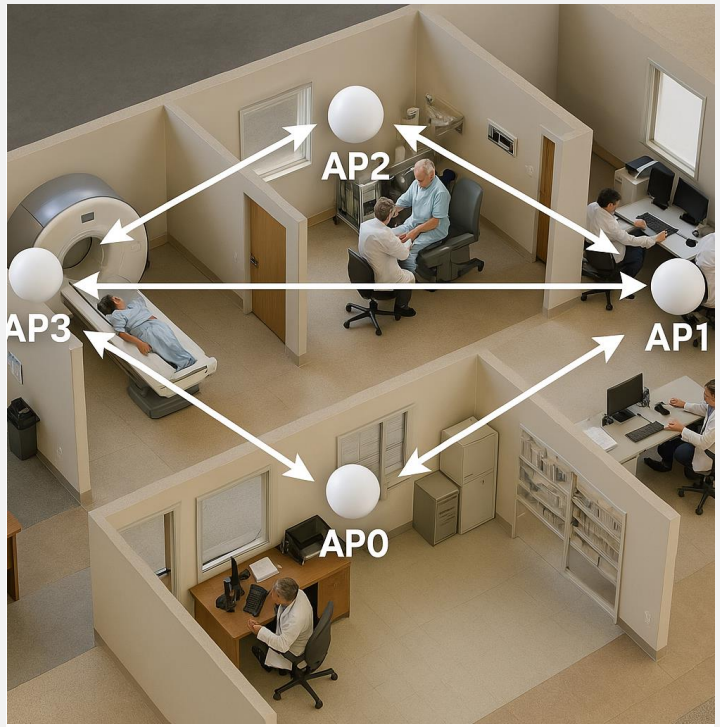
Geofencing - security alerts

LOCALIZATION

Indoor asset management - hospitals, warehouses

Pet tracking inside home

Item finding - wallet, keys



AUTOMAPPING

Solar Trackers

Luminaires, Access Points

Accurate Mapping for Battery Storage

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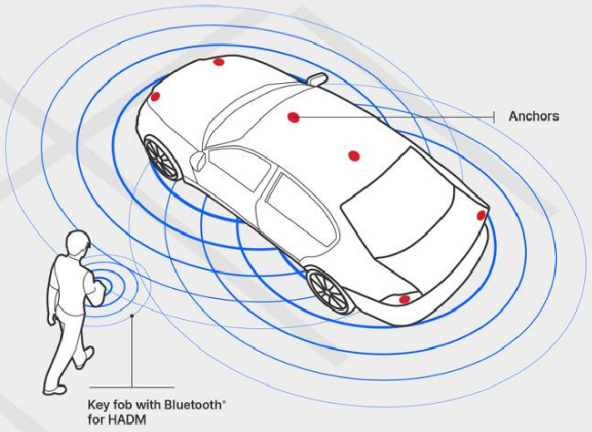
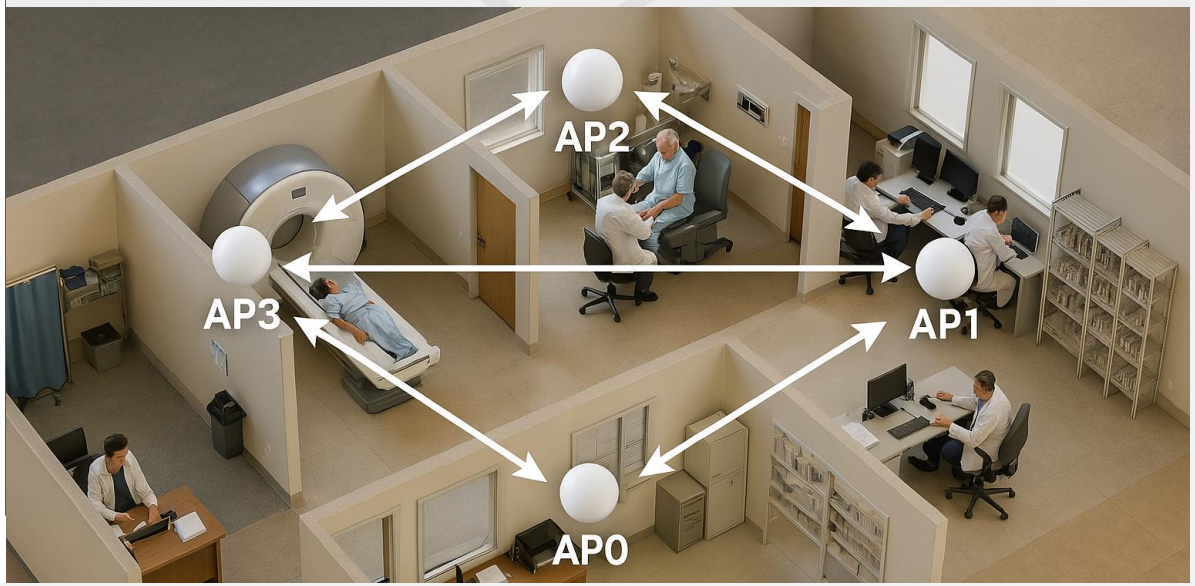
Channel Sounding: Real-World Applications and Strategic Differentiators

- **Enables a wide range of spatially-aware IoT applications**
  - **Keyless Entry (PEPS)** — Enables zonal detection and secure access
  - **Static Device Mapping** — Precise positioning of luminaires, access points
  - **Asset Tracking** — Equipment in hospitals, factories, or warehouses
  - **Retail Loss Prevention** — Detects unauthorized item movement
  - **Geofencing** — Triggers actions based on movement within set boundaries
  - **Item Finding** — Enables precise tracking of personal belongings



- **A Differentiated, End-to-End Channel Sounding Solution**
  - **In-house IP** — Optimized, royalty-free, and stack ready
  - **End-to-End Solution** — GA hardware, software, tools, in one package
  - **Antenna Flexibility** — Single or multi-antenna for optimal trade-offs
  - **Compact Dev Kit** — Ideal for size-constrained applications like key fobs



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Channel Sounding in an Indoor Facility

Access Control

- Restrict access to unauthorized personnel
- Send alerts to local servers/cloud if anyone dwells in the area for too long

Entry access

Authenticate and grant access to authorized workers when they approach the door

Asset management

- Coarse localization of inventory inside the facility
- Increase worker efficiency

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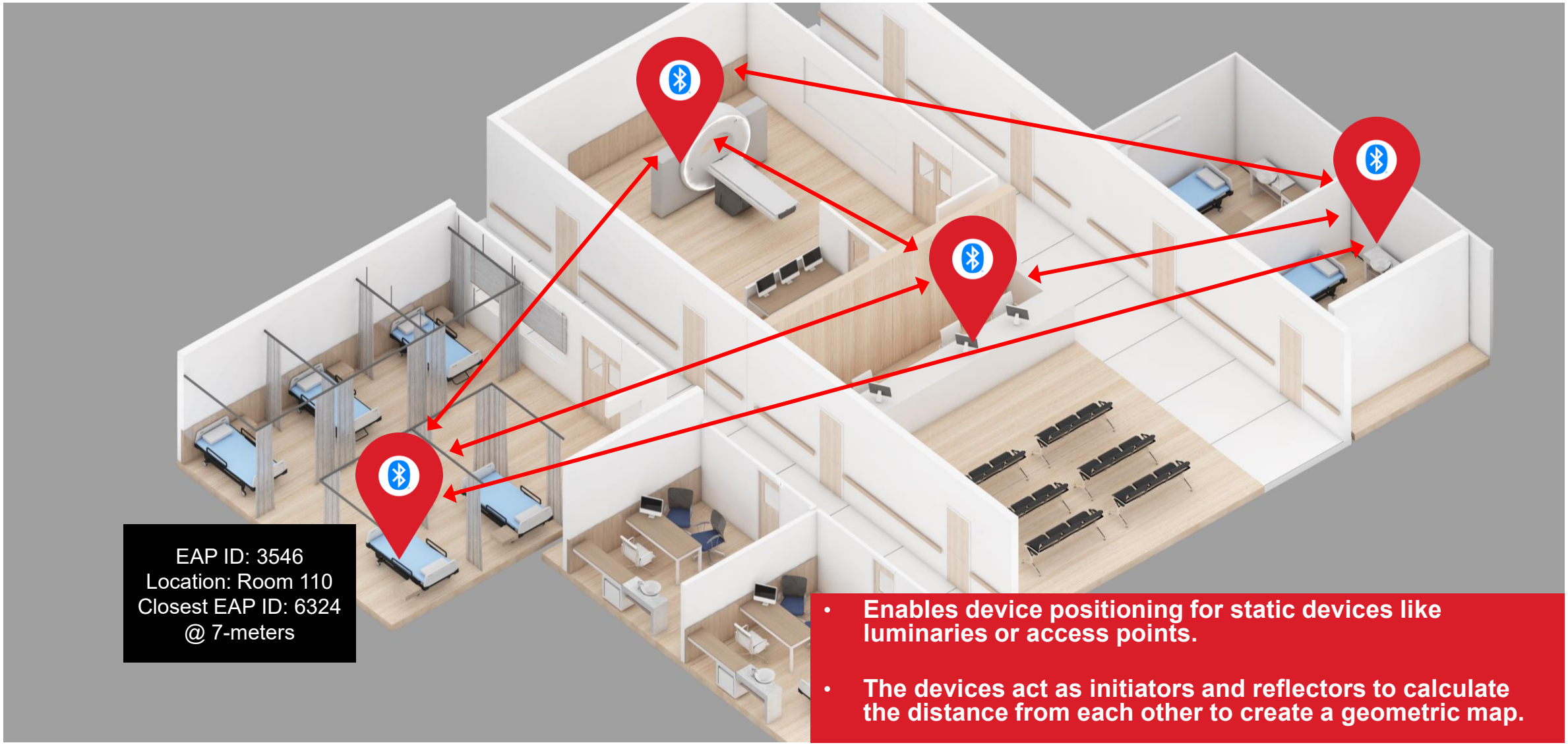
Channel Sounding for Static Device Positioning



- **Enables device positioning for static devices like luminaries or access points.**
- **The devices act as initiators and reflectors to calculate the distance from each other to create a geometric map.**

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Channel Sounding Overview

- **Measure distance between two devices using**
  - Phase-based Ranging (PBR)
  - Round Trip Time (RTT)
- **RTT and PBR operates across 2.4 GHz band**
  - Standard specifies up to 72 channels
  - Random hopping pattern
- **Connection-Oriented 2-way ranging with two roles**
  - Initiator: device that wishes to calculate distance from itself to another device
  - Reflector: device responding to initiator
- **Supports up to 4 antenna paths between devices**
  - 8 possible antenna combinations
- **Multiple security features included in the standard**
- **Can be combined with Angle of Arrival / Departure (AoA/AoD)**
  - Enables position estimation with single initiator/reflector pair

Additional Resources

- Webpage – Learn more about Silicon Labs offerings and demos
- Tech Talk – Explore Bluetooth Channel Sounding
- Workswith 2024 - Enable Accurate Distance Estimation Using Channel Sounding
- Blog: Learn more about Antenna Switching with Silicon Labs Channel Sounding
- API Spec: Getting Started with Silicon Labs Bluetooth Channel Sounding

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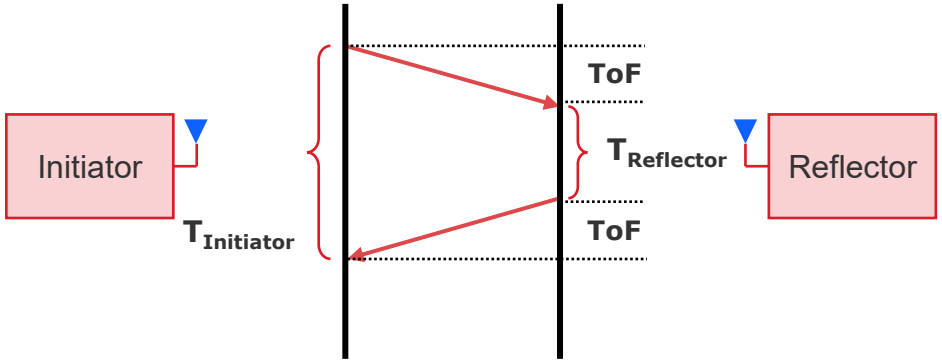
Channel Sounding Step Modes

Mode-0: Calibration

- Compensates for clock drift and frequency offset
- Results in fractional frequency offset table

Required

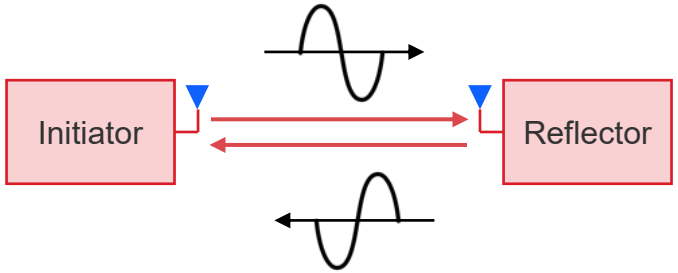
Mode-1: Round trip time



- CS SYNC packets exchanged between initiator and reflector

Required

Mode-2: Phase based ranging



- CS Tone exchanged between initiator and reflector
- Each antenna path exercised in each step

Required

Mode-3: PBR+RTT

- Combined PBR and RTT in each step
- RTT distance measurement can be cross-checked with PBR results
- Provides higher security as a mismatch in distance estimation can indicate relay attack

Optional

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Bluetooth® LE Location Services Comparison

	RSSI	UWB	Angle of Arrival	Channel Sounding
Localization metric	Resolve distance estimation from transmitter signal strength	Resolve distance between two points using high-resolution time-of-flight (ToF)	Resolve relative angle between two points	Resolve distance between two points using time of flight and phase-based ranging
Antenna requirements	Single antenna	Multiple antennas; often complex hardware	Multi-antenna required by spec	Single or dual antenna
Connectivity	Connection-oriented and connectionless	Connectionless	Connection-oriented and connectionless	Connection-oriented
Performance metrics	+/- 5 m, high susceptibility to multipath interference	High accuracy (~10–30 cm)	+- 3 degrees accuracy – azimuth +- 5 degrees accuracy – elevation	+- .3 m < 5m with PBR +- 0.5 m > 5m with PBR
Solution advantages	Ubiquitous support for RSSI measurements in existing Bluetooth LE products	• High Precision • Low Latency	• Scalable solution for real time position tracking • Supports 5-10 year battery life	• Small form factor with flexible antenna design • Feature-add for security by proximity
Solution disadvantages	Highly susceptible to RF noise and multipath	High power usage; expensive and complex integration	• Needs complex antenna setup and calibration	Scalability

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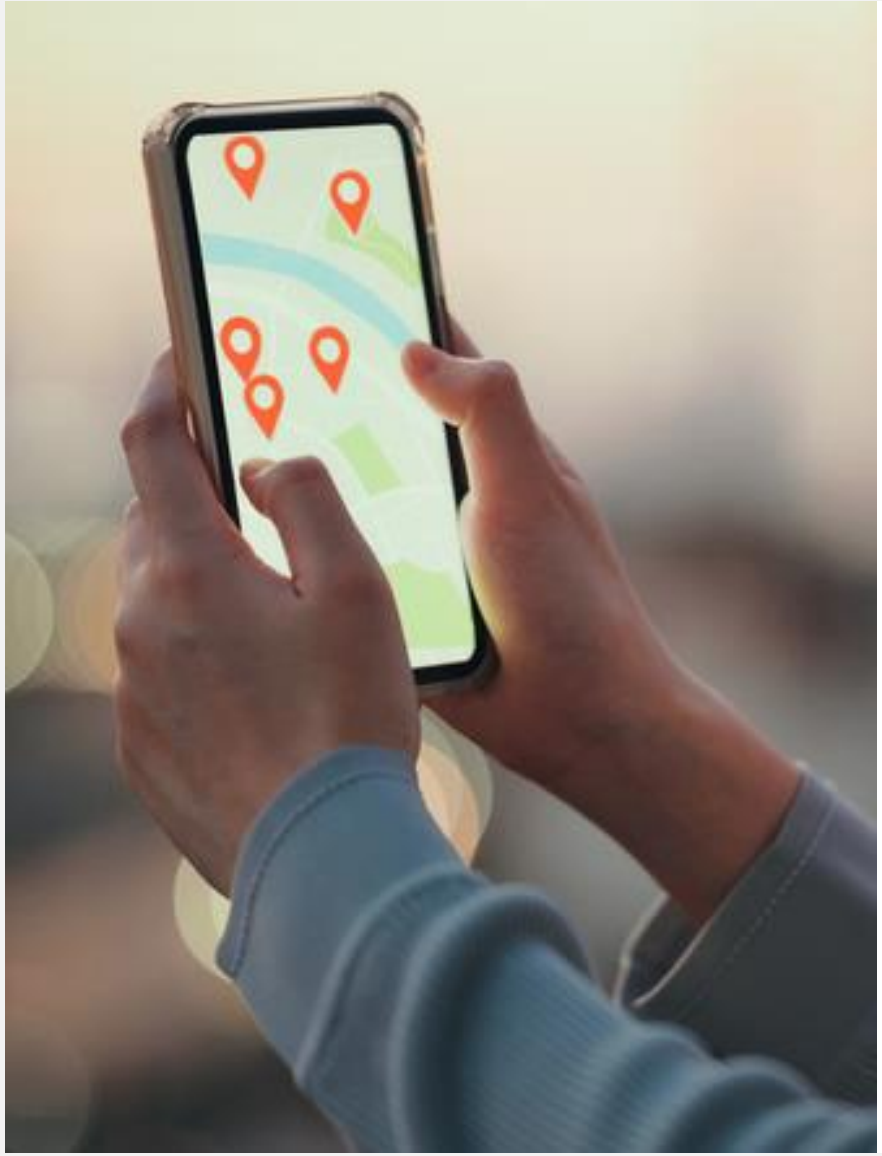
Android Support for Channel Sounding



- **Android**
  - Channel Sounding API support available since Android 16
  - Google Pixel 10 is first Android device with official Channel Sounding support
  - Si Connect with Channel Sounding demo
  - Expect enablement of Channel Sounding in Google FindHub ecosystem in early 2026

25Q4	26Q2
Google Pixel 10 as Central, Initiator ↕ xG24-2606A as Peripheral, Reflector	Google Pixel 10 as Central, Reflector ↕ xG24-2606A as Peripheral, Initiator
Google's distance estimation algorithm	Silicon Labs distance estimation algorithm
Use-case: Find Hub, asset tracking/locationing	Use-case: Phone-as-digital-key, access control, door locks, passive entry for automotive, and so on.

- **Get Started:** https://github.com/SiliconLabs/SimplicityConnect-android

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Ranging Modes - Phone as Initiator vs Reflector

	Phone as Initiator	Phone as reflector (Embedded Initiator)
Overview	Phone initiates ranging; embedded device reflects. Ideal for tracking (e.g. FindHub)	Embedded device (DK2606) initiates; phone reflects. Ideal for access control applications.
Algorithm	Uses Google's Channel Sounding algorithm, limited to Android ecosystem.	Uses Silicon Labs Channel Sounding algorithm, enabling configurable filtering and real-time algorithm modes.
Product Support	xG24, BG24L BG24L usable as reflector (no extra compute/memory needed)	xG24 only Requires higher compute/memory (not supported on BG24L)
Use Cases	Find Hub, asset tracking & locationing	Phone-as-digital-key, access control, door locks, passive entry for automotive, and so on.

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Performance in Indoor Office Environment

- **Ceiling rail infrastructure**

  Internal test environment

  Multiple stationary EFR32 devices placed at different locations

  Mobile EFR32 device for controlled measurements (repeatability)

- **Challenges - heavy multi-path in an indoor office setting**

- **Statistical analysis**

  Static measurements at multiple distances up to 33 meters

  Hundreds of measurements per distance to determine min/max, mean, median, std, absolute error

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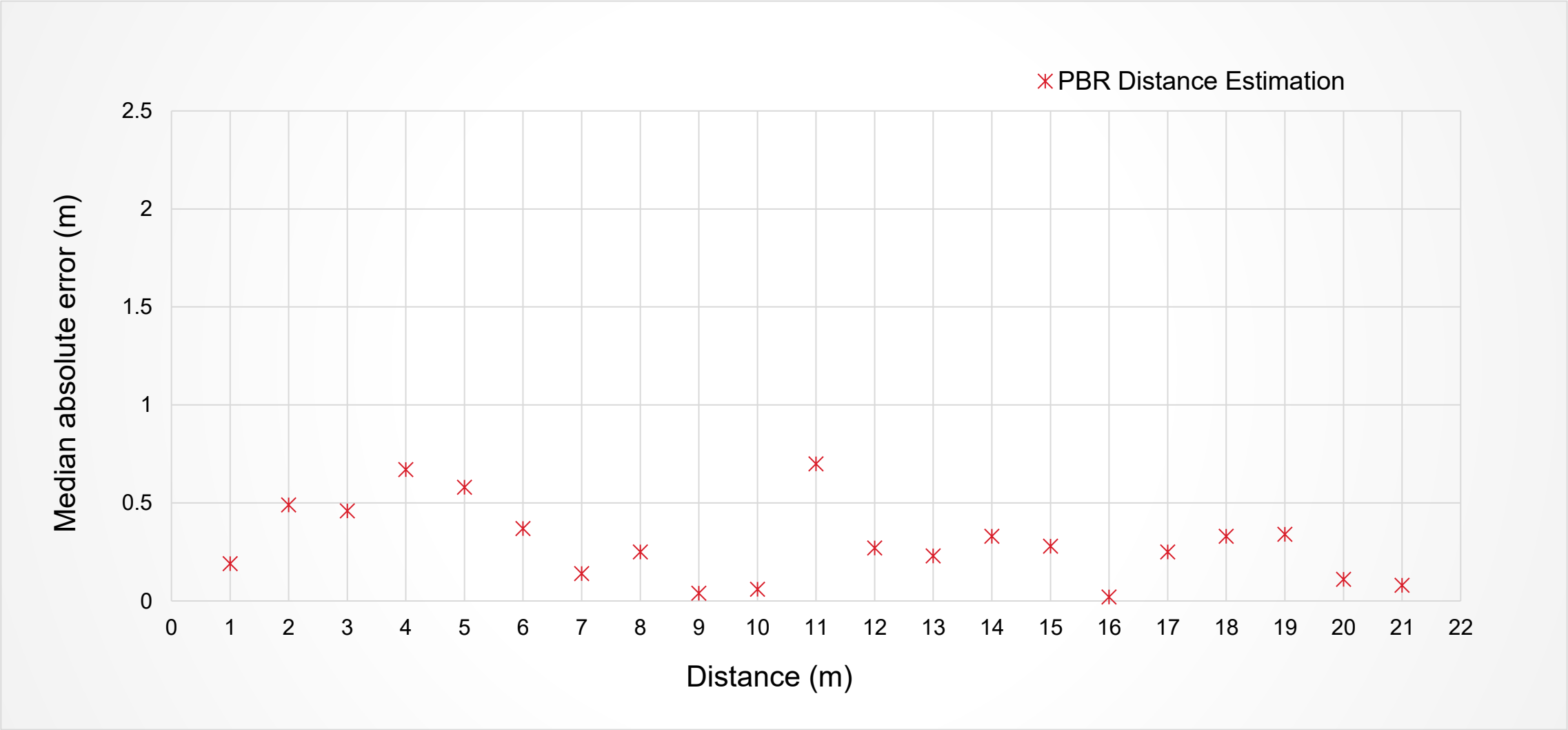
Indoor Performance Result – Line of Sight



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Single vs Dual Antenna

Board	Environment	90th Percentile Absolute Error	95th Percentile Absolute Error	Std. deviation	Mean signed error (in m)
BRD4198A (1x1 = 1 antenna path, 72 channels)	Line of sight	0.57	0.58	0.24	0.21
	Non-line of sight	3.79	4.56	1.79	1.38
BRD2606A (2x2 = 4 antenna paths, 72 channels)	Line of sight	0.57	0.69	0.2	-0.41
	Non-line of sight	1.91	4.07	1.64	0.89

Antenna Diversity Substantially Improves Distance Estimation Accuracy Performance in Non-line of Sight Scenarios

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Single vs Dual Antenna

Board	Number of Channel Sounding Channels[1]	Algorithm computation time[1] (ms)	Measurement Update Frequency[1] (Hz)	Total Energy[2] (nAh)	Theoretical Measurement Range (m)
BRD4198A (1x1 = 1 antenna path)	72	~43.4	~6.1	124	150
	37	~12.6	~8.9	73	75
BRD2606A (2 x 2 = 4 antenna paths)	72	~189	~3.05	222	150
	37	~47	~6.3	116	75

1 Measured per measurement at Initiator with algorithm mode SL_RTL_ALGO_MODE_REAL_TIME_BASIC
2 Measured per measurement at Reflector

Antenna Diversity Increases Measurement Latency and Total Energy Per Measurement

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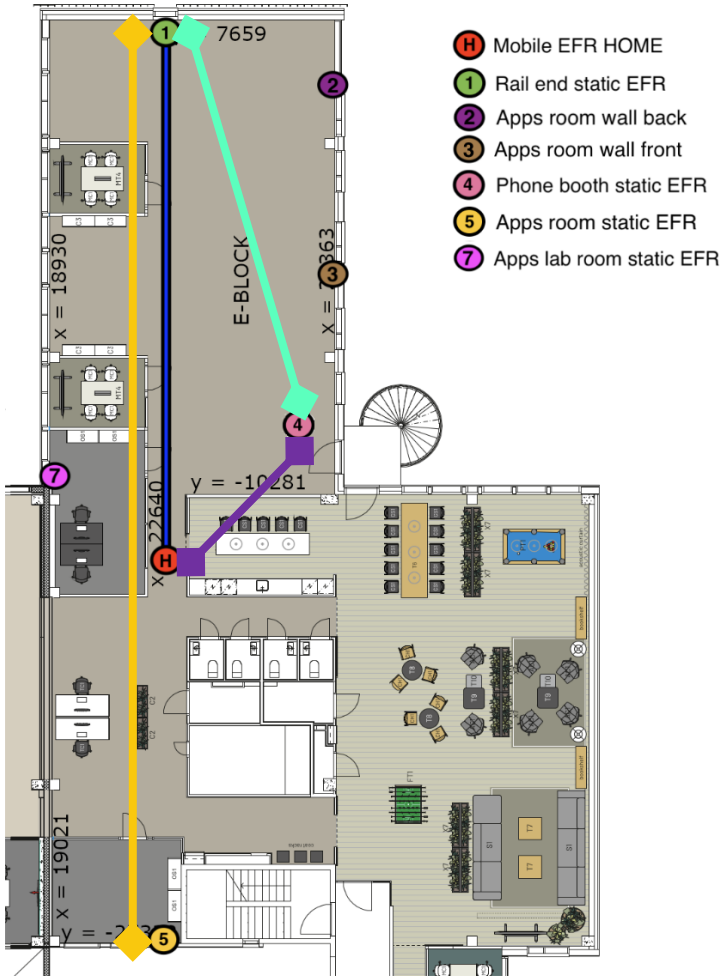
Sub-meter accuracy for NLOS (Non-Line-of-Sight)



Nodes	Distance	Obstacles	Error
1 & 5	33.9m	Walls	+/- 0.3 m
1 & 4	14.84m	Cubicles	+/- 0.2 m
H & 4	7.9m	Walls, Kitchen	+/- 0.5 m

- **NLOS performance indoors**
  - Setup in office space with nodes on the ceiling
  - Nodes face various obstacles: walls, kitchen, office cubicles
  - Utilizes antenna diversity and intra-event antenna switching
  - Dual-antenna solution ensures robust accuracy

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BG24: Optimized for Battery Powered, Channel Sounding-enabled IoT Devices

Bluetooth® Proprietary

- **5x5 QFN40 (26 GPIO), AEC-Q100**
- **6x6 QFN48 (32 GPIO), AEC-Q100**
- **3.1x3.0 WLCSP42**

DIFFERENTIATED FEATURES

- **Ultra small form-factor**
  - 3.1 x 3.0 WLCSP package
- **+20 dBm output power**
  - Eliminates need for external power amplify
- **AI/ML accelerator**
  - Accelerates inferencing while reducing power consumption
- **Secure Vault High**
  - Protects data and device from local and remote attacks
- **20-bit ADC**
  - 16-bit ENOB for advance sensing
- **Improved Coexistence**
  - Ideal for gateways and hubs
- **PLFRCO**
  - Eliminates need for 32 KHz xtal

DEVICE SPECIFICATIONS

- **High Performance Radio**
  - Up to +19.5 dBm TX
  - -97.6 dBm RX @ BLE 1 Mbps
- **Efficient ARM® Cortex®-M33**
  - Up to 78 MHz
  - 1536kB Flash, 256kB RAM
- **Low Power**
  - 49.1 µA/MHz (CoreMark)
  - 5.0 mA TX @ 0 dBm
  - 5.1 mA RX (802.15.4)
  - 4.4 mA RX (BLE 1 Mbps)
  - 1.3 µA EM2 sleep
- **Multiple protocol support**
  - Bluetooth 6.0 (1M/2M/LR), Bluetooth mesh, Proprietary 2.4 GHz

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BG24L: Channel Sounding Optimized, High-Performance & Low-Cost AI/ML Wireless SoC

Bluetooth®

- **5x5 QFN40 (26 GPIO)**

DIFFERENTIATED FEATURES

- **Supports Bluetooth 6.0**
  - Channel Sounding optimized BLE SoC
  - Single-connection two-way ranging
  - Ideal Solution for Channel Sounding tags
- **Lowest Power RF**
  - Increases battery life
- **PLFRCO**
  - Eliminates need for 32 KHz XTAL and lowers overall system cost
- **16-bit ADC**
  - Up to 14-bit ENOB for better analog sensing
- **AI/ML accelerator**
  - Accelerates inferencing while reducing power consumption
- **Secure Vault Mid**
  - Protects data and device from local and remote attacks
- **Improved Coexistence**
  - Ideal for gateways and hubs

DEVICE SPECIFICATIONS

- **High Performance Radio**
  - Up to +10 dBm TX
  - -97.6 dBm RX @ BLE 1 Mbps
- **Efficient ARM® Cortex®-M33**
  - Up to 78 MHz
  - 768kB Flash, 96kB RAM
- **Low Power**
  - 49.1 µA/MHz (CoreMark)
  - 5.0 mA TX @ 0 dBm
  - 5.1 mA RX (802.15.4)
  - 4.4 mA RX (BLE 1 Mbps)
  - 1.3 µA EM2 (16kB RAM retention)
- **Wide Operating Range**
  - 1.71 to 3.8 volts
  - +125ºC operating temperature
- **Multiple protocol support**
  - Bluetooth 6.0 (1M/2M/LR), Bluetooth mesh, Proprietary 2.4 GHz

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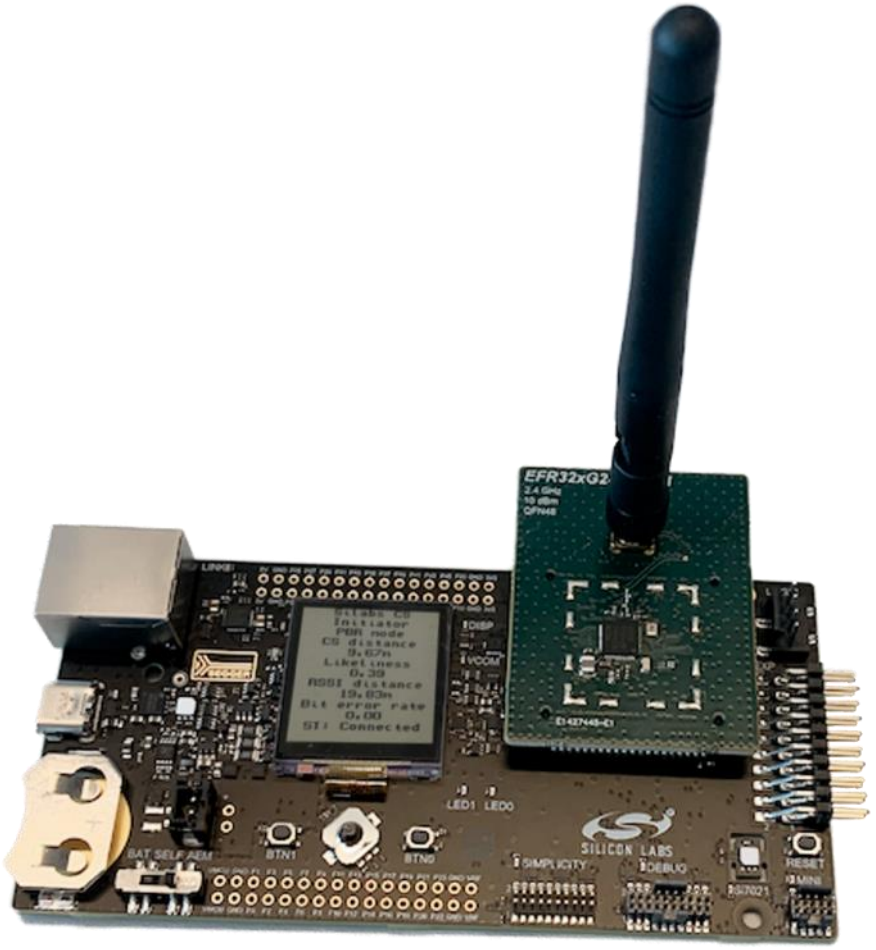
Single-Antenna Channel Sounding Board



- **Channel Sounding evaluation using Ranging Kit**
  - EFR32xG24 Pro Kit with external antenna
  - Full featured development option leveraging WPK features
  - AEC-Q100 Compliant
  - SoC/NCP Sample Apps
    - Initiator and Reflector examples supported
  - Ranging Library
    - Process IQ samples, post-filtering, and compute distance using configurable algorithm
- **Optimized antenna designs**
  - EFR32xG24 reference designs with optimized PCB antenna solutions for indoor location systems

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Dual-Antenna Channel Sounding board

- **xG24 Channel Sounding Development Kit**
  - Available since March 2025
  - Development Kit with two PCB antennas
    - Antenna diversity offers increased robustness and accuracy
  - Intra-event antenna switching for optimal non-line of sight performance
  - Includes IMU sensor to detect movement & wake-up the tag
  - Small form factor
    - Ideal for size-constrained applications like key fobs
  - AEC-Q100 Compliant
  - SoC/NCP Sample Apps
    - Initiator and Reflector examples supported
  - Ranging Library
    - Process IQ samples, post-filtering, and compute distance using configurable algorithm

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End-to-End Channel Sounding Solution: Integrated, In-House, Royalty-Free

- **Channel Sounding Algorithm Features**
  - Supports Multiple Channel Sounding Ranging Modes
    - PBR, RTT, PBR with RTT as submode
  - Antenna Switching
    - Built-in support for antenna diversity. Supports 1, 2 and 4 antenna paths
  - Supported Algorithm modes
    - Static mode – High accuracy
    - Real Time Basic – Offers high accuracy with increased computation and measurement latency; supports tracking speeds up to 1 m/s
    - Real Time Fast – Offers moderate accuracy with lower computation and measurement latency, supports tracking up to 2.1 m/s
  - Configurable Channel Selection (72, 37, or 20 Channels)
    - Selectable based on accuracy needs and power constraints
  - Integrated Real-time visualization tool with Simplicity Studio
    - Object tracking – Displays accurate range to the target device
    - Relative Radial Velocity metric – Calculates how quickly devices move toward or away from each other

- **Key Benefits Of Our In-House Solution**
  - Licensing-Free Deployment
    - Eliminates third-party royalties, simplifying BOM cost structure and legal risk
  - Optimized HW-SW Co-Design
    - Tight coupling between silicon & firmware ensures seamless performance & efficiency
  - Single-Vendor Lifecycle Support
    - Unified hardware & software ownership streamlines debugging, validation, and updates

Feature	Silicon Labs	Closest Competitor
Algorithm IP	✅ In-house developed	❌ Licensed 3rd-Party
Software Stack	✅ Integrated FW + SW	⚠️ Layered Integration with External IP Blocks
IP Licensing Model	✅ Fully Owned Stack, Royalty-Free	❌ Perpetual or Usage-Based Licensing Fees
Vendor Support	✅ Unified Silicon + Stack Support	⚠️ Fragmented Support Across Vendors

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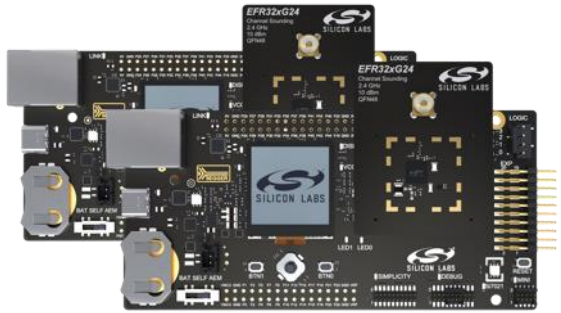
Getting Started with Channel Sounding

- **Dev Board**
  - 1x Development board
    - Dual PCB antennas
    - On-board debugger
    - Signal breakouts
    - On-board sensors
    - 20-bit ADC
    - AI/ML hardware accelerator

Part Number	Description
xG24-DK2606A	xG24 +10dBm Channel Sounding Dev Kit

- **Pro kits**
  - 2x radio board
  - 2x WSTK main board
    - Modular development platform
    - Advanced development
    - RF measurements
    - Energy profiling
    - External device debug
    - Ethernet for large network test
  - 2x Sleeve Dipole Antenna



Part Number	Description
xG24-PK6036A	xG24 +10dBm Channel Sounding Pro Kit

- **Radio Board kits**
  - 1x radio board
    - Uses existing WSTK boards
    - Uses existing software tools
  - 1x Sleeve Dipole Antenna

Part Number	Description
XG24-RB4198A	6x6QFN48, +10dBm Channel Sounding Radio Board

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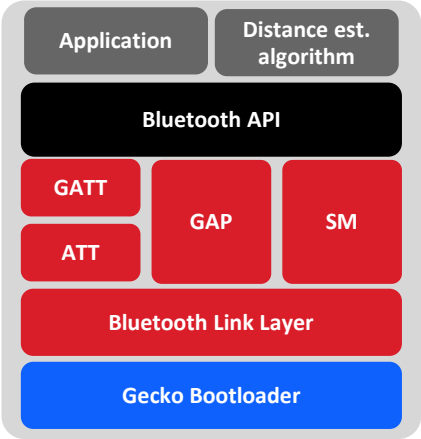
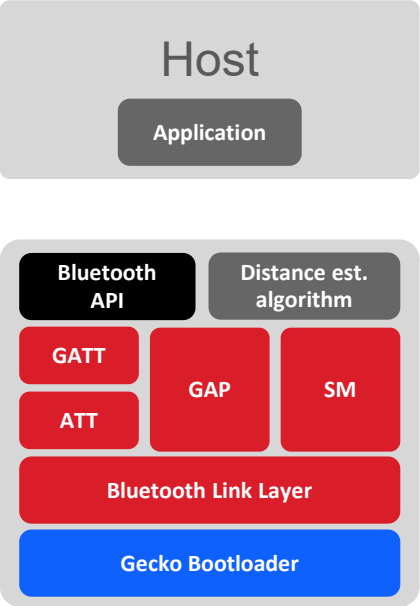
Bluetooth Software for Channel Sounding



SoC



NCP

- **Bluetooth 6.0 support for Channel Sounding**
  - All mandatory Channel Sounding features supported
  - Distance estimation algorithm included
- **SoC and NCP modes**
  - Channel Sounding data over BG API and distance estimation calculated on-chip
- **Software components included in Simplicity Studio**
- **Example applications available**
- **Typical Channel Sounding application with stacks is 500K**
  - Leaves 60%+ for user applications
- **Documentation: http://docs.silabs.com/**

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Visualizer Tool

Visualizer Tool displays realtime data

- Channel sounding data with RSSI readings for comparison
- Interfaces with Channel Sounding-enabled EVKs

Upcoming features

- Data logging
- Confidence metric display
- Channel map selection

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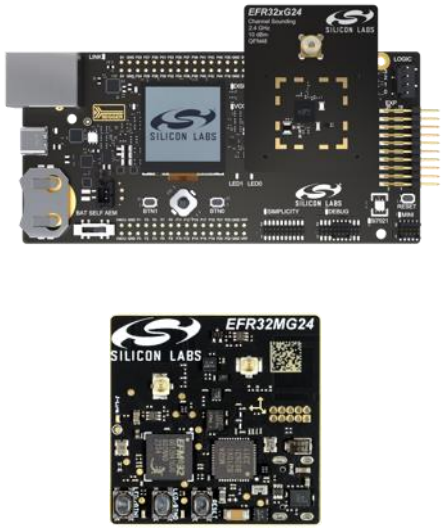
Silicon Labs Bluetooth ® Channel Sounding Offering



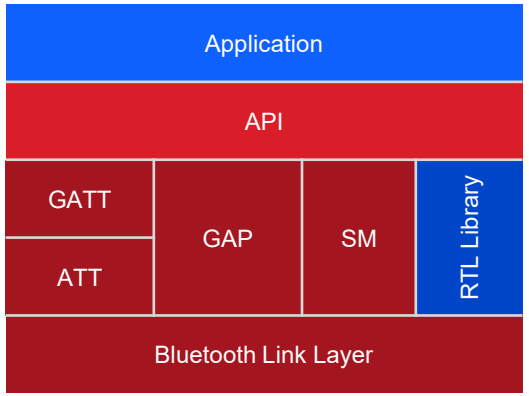
ICS & DEVELOPMENT KITS

Channel Sounding Supported by **B/MG24**

Kits:

xG24-RB4198A single antenna kit

xG24-DK2606A dual antenna kit

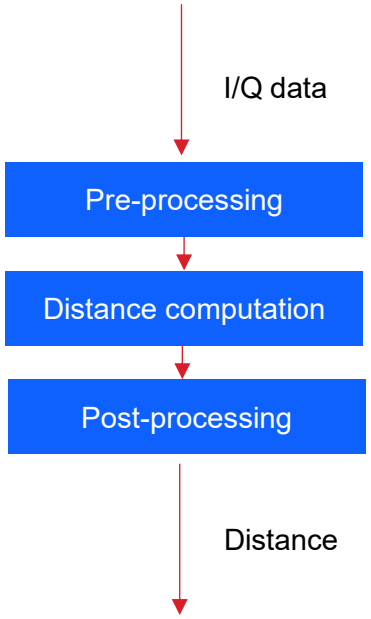


BLUETOOTH 6.0 STACK

In-house developed stack, supported and maintained stack

Bluetooth 6.0 qualified

PBR & RTT Modes



RTL LIBRARY

Computes distance from raw I/Q data

Developed and supported by Silicon Labs

New features added based on market needs

No 3rd party license fees

SDK & TOOLS

Initiator & Reflector examples

Real-time visualization tool for Bluetooth Channel Sounding

Energy Profiler etc.

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Channel Sounding on BG24L and BG24

Application Type	OTA	BG24	BG24L
SoC Reflector	Application based OTA	Yes	Yes
SoC Initiator	Application based OTA	Yes	No
NCP Initiator + Reflector	BGAPI UART OTA	Yes	No

BG24L Suitable for Simple Tags with Channel Sounding Support

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Channel Sounding – Revolutionizing Secure, High-Accuracy Distance Measurement

HIGH ACCURACY DISTANCE ESTIMATION

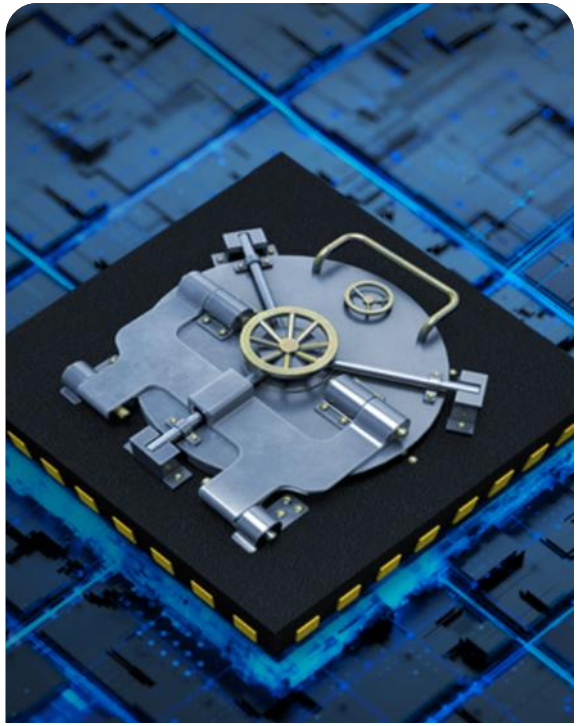
- Distance estimation using PBR and RTT
- Support for Time of Flight and Multi-antenna support for optimized non-line-of-sight ranging

INTEROPERABILITY AND RELIABILITY

- Compatible with a wide range of devices
- The device performance is ensured through Bluetooth qualification.

COST EFFECTIVE AND EFFICIENT

- Simple layout design which saves board space and cost
- Unlike other solutions like UWB, Channel Sounding does not require additional components.



ENHANCED SECURITY

- Reduces likelihood of man-in-middle and spoofing attacks
- Higher security through Deterministic Random Bit Generator

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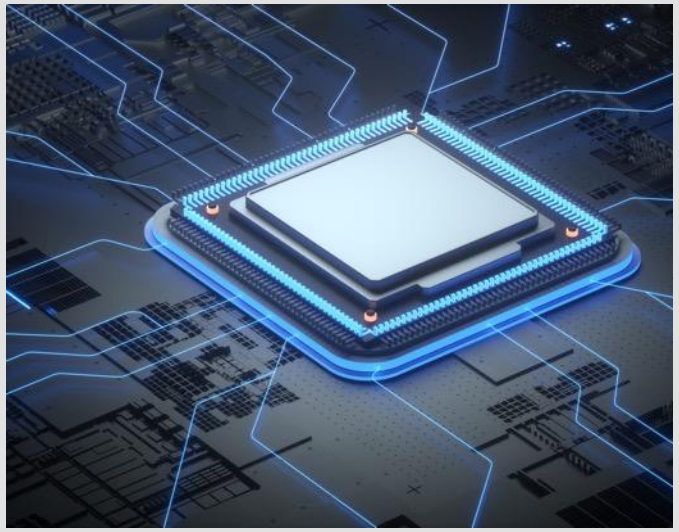
Learn More About Silicon Labs Channel Sounding



GETTING STARTED

Channel Sounding Kits Available at:

Visit site

Download Simplicity Studio here:

Visit site

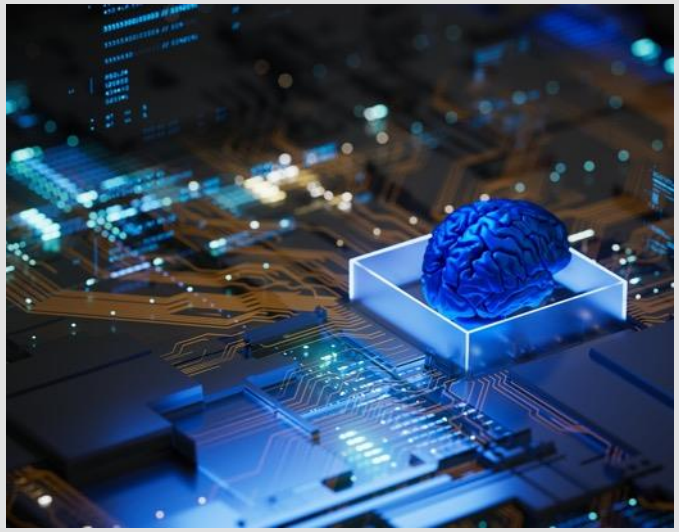
FOR MORE INFORMATION

Channel Sounding:

Visit site

Explore Bluetooth Channel Sounding Webinar:

Visit site



CALL TO ACTION

For any questions about SiLabs offerings, please contact our Sales team.

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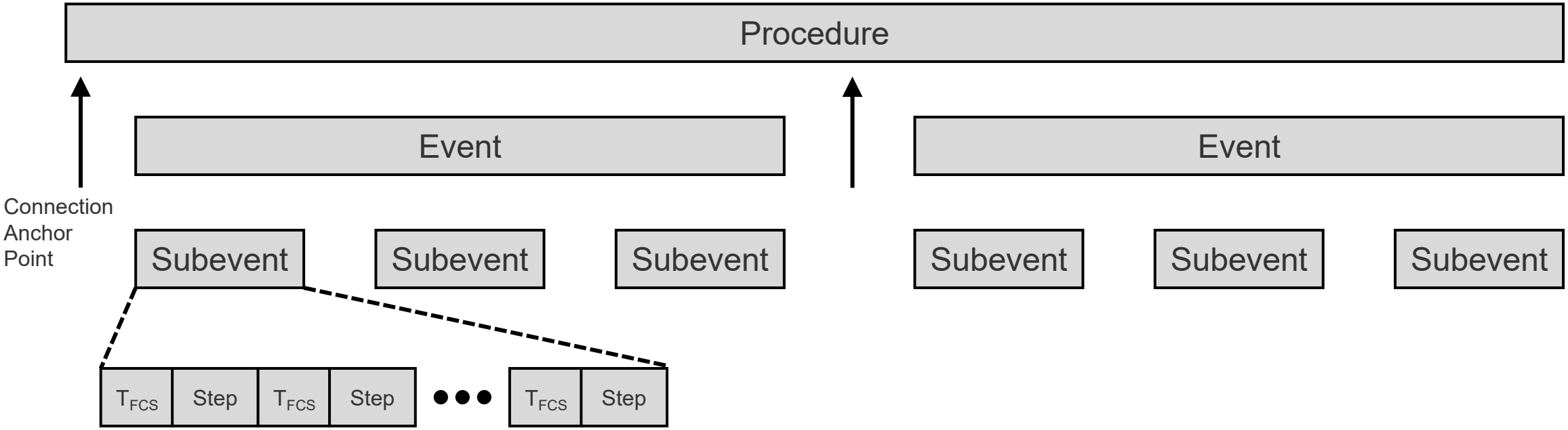
Thank you

Back-up

Procedure / Events / Subevents / Steps



- Devices exchange Channel Sounding(CS) capabilities and security parameters before starting a procedure
- Subevent parameters such as Channel Sounding sequence, antenna switching pattern, access address etc. are random and are based on the initial parameters exchanged between the devices

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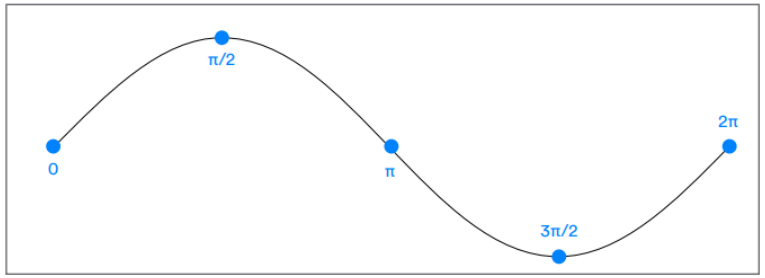
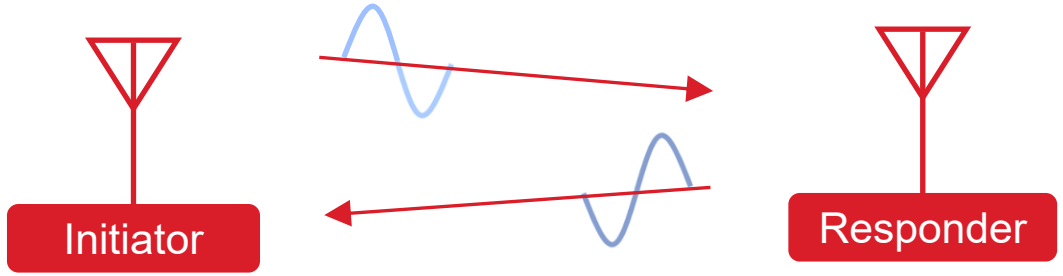
Phase-Based Ranging (PBR) Measurement

Phase
A specific point in a wave cycle, perhaps measured as the wave passes over an antenna, is known as its *phase*. Phase is measured as an angle from 0 at the start of the wave cycle to 360 degrees or 2π radians at the end of the wave cycle.





- **Phase of RF signals is a function of frequency of the carrier and the distance traveled**
  - Phase rotation due to spatial propagation desired
  - Measurements at multiple RF frequencies to resolve distance ambiguity
- **Distance is calculated using the phase difference between transmitted and received signal**
- **Distance measurement process**
  - Calibration phase
  - Measurement phase where both devices transmits a packet and calculates the phase difference
  - Signal processing and distance calculation algorithm
- **Security benefits**
  - Difficult for an attacker to manipulate phase
  - Random transmission length and channel map makes sniffing difficult
- **0.3 m accuracy**

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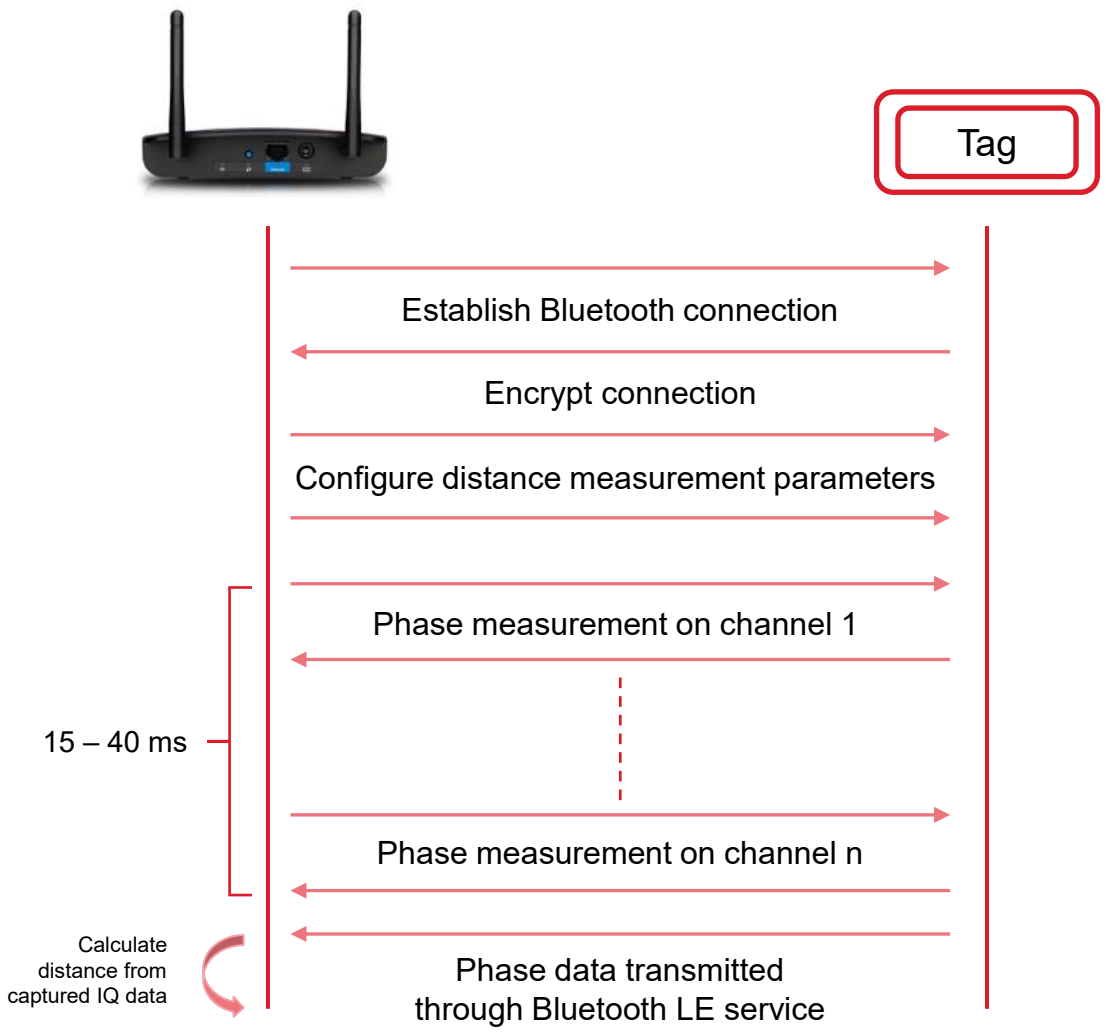
Example PBR Procedure within Bluetooth LE Connection



- **Connection-based two-way ranging with secure Bluetooth LE connection**
- **Configurable distance calculation algorithm parameters**
  - Number of channels and channel map
  - Phase transmission duration
  - Trade-offs between accuracy, duration, and power
- **Access points and tags transmit signal to measure phase on both devices**
  - The ranging algorithm in the AP processes the IQ data and calculates the distance using phase difference and measurements at multiple channels

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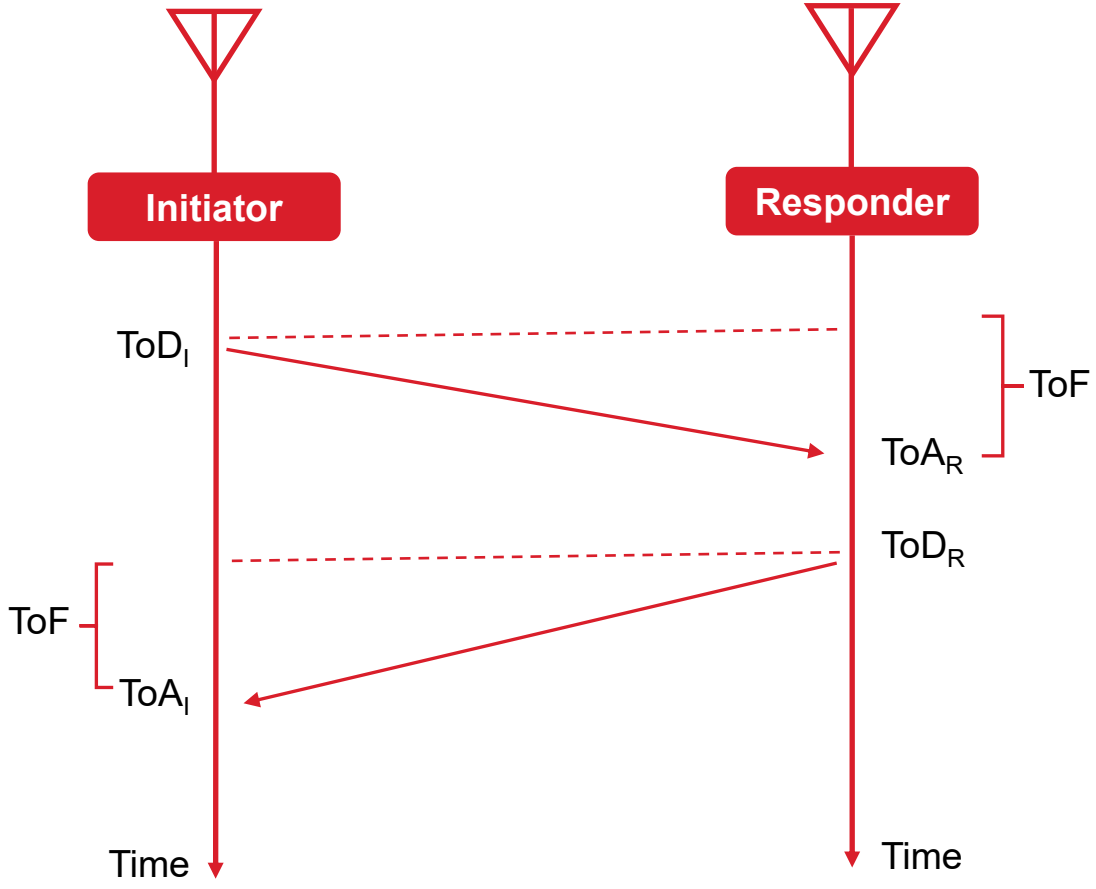
Round Trip Time (RTT) Measurement



$$RTT = 2\,ToF = (ToA_I - ToD_I) - (ToD_R - ToA_R)$$

- **Time of Flight (ToF) is measured on both initiator and reflector side using Time-of-Arrival (ToA) and Time-of-Departure (ToD)**
  - Distance is estimated from exchanged measurements on multiple channels
  - Fractional techniques used to resolve sampling uncertainty and increase accuracy
- **Duration of measurement procedure is variable**
  - Procedures can be split across multiple intervals
  - Multiple measurements in longer connection interval
  - Number of channels
- **Security benefits**
  - Time cannot be reversed, preventing reflector from appearing closer than it is
  - Random bits as sync word guard against packet sniffing
  - Random tone length prevents attackers from altering toning sequence
- **1-5m accuracy**

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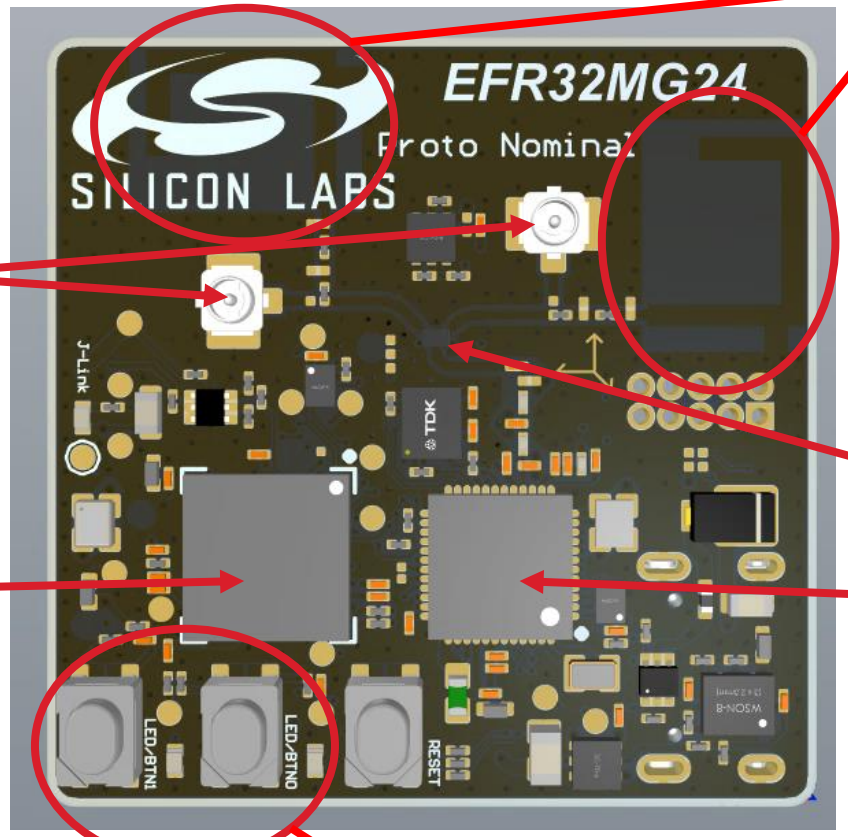
Dual-Antenna Channel Sounding board - front



2x PCB antenna for antenna diversity and dual polarization

2x U.FL connector for optional external antenna support

1-to-2 RF switch

On-board debugger for easy programming and debugging

EFR32MG24B210F1536IM48-B
our most capable xG24 part
(with 1.5MB flash, Channel Sounding, MVP, Secure Vault High, +10dBm)

User buttons and LEDs
for demo firmware support

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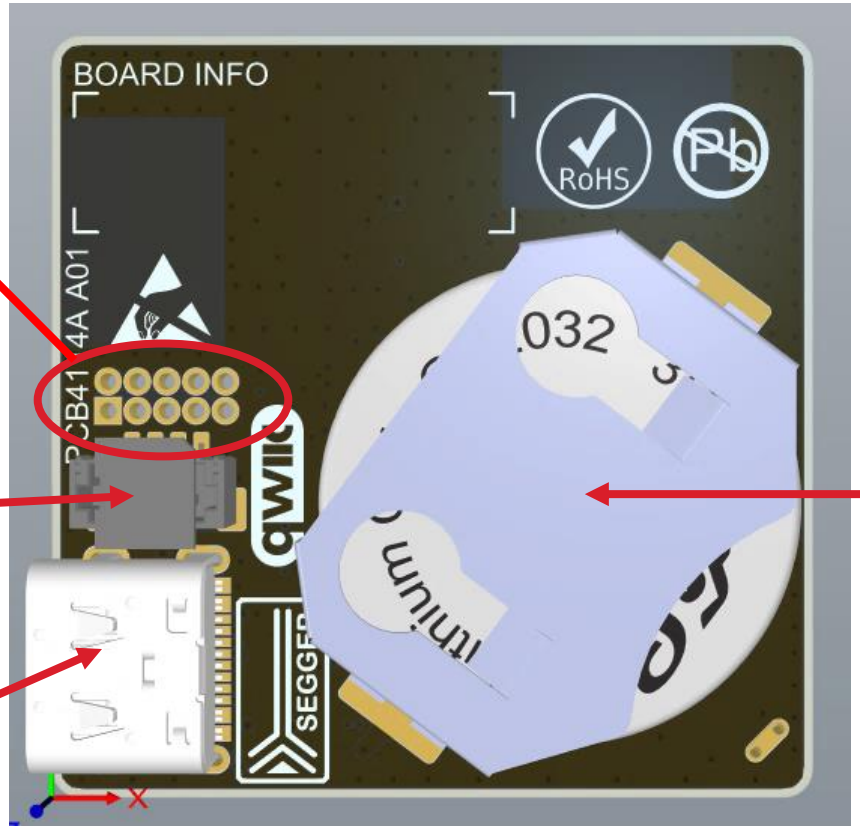
Dual antenna Channel Sounding board - back

33x33mm overall size

Mini Simplicity header for Wireless Main Board connectivity. Provides Advanced Energy Monitoring and Packet Trace Interface options.



Qwiic connector for extendibility

CR2032 for portability

USB-C connector for easy PC connectivity. Provides both debugger and virtual UART interfaces

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